

A Template to be used by Students for Typesetting FYDP Synopsis

A Synopsis Submitted in Partial Fulfillment of the Requirements
for the Degree of Bachelor of Science in Electrical Engineering

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Sustainable Development Goals

1. No poverty
2. Zero hunger
3. Good health and well-being
4. Quality education
5. Clean water and sanitation
6. Affordable and clean energy
7. Decent work and economic growth
8. Industry, innovation and infrastructure
9. Sustainable cities and communities
10. Responsible consumption and production
11. Climate action
12. Life below water
13. Life on land
14. Peace, justice and strong institutions

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Abbreviations

LAH List Abbreviations **Here**

Abstract

This section should be a complete, self-contained summary of your entire project proposal. It must be written after you have completed all the other sections. Briefly cover the context of the project, the core problem you are addressing, your key objectives, the proposed methodology, and the expected outcomes. It gives the reader a quick overview of the entire synopsis. (Maximum 350 words)

Contribution to Sustainable Development Goals

Write a comprehensive description of how the work proposed in this synopsis aims at contributing to one or more of the sustainable development goals listed above on page i.

Chapter 1

Introduction

Start with a brief overview of the broader domain your project falls into (e.g., Renewable Energy, Power Systems, Embedded Systems, Robotics, etc.). Explain the context and importance of this area. Then, introduce the specific problem or technology your project will address. Clearly state the motivation behind your project and what you aim to achieve in a general sense. (1 – 2 pages)

Chapter 2

Problem Statement

Clearly and concisely define the specific problem your project intends to solve. This section should be direct and to the point. Explain the gap in current technology or the specific challenge you are addressing. (1 page)

Chapter 3

Literature Review

Summarize your background research on existing solutions and technologies. Discuss findings from (at least 10) key research papers, books and/or articles, highlighting their limitations and how your project will improve upon them. This section demonstrates that you have studied the field and your project is building upon existing knowledge. Cite your sources in the text using numbers in square brackets [1]. (2 – 3 pages)

Chapter 4

Objectives

List the specific, measurable, achievable, relevant, and time-bound (SMART) objectives of your project. Use a numbered or bulleted list.

Discuss how is your solution going to be different and better than others?

Students must describe the final project output in detail, its expected packaging, and hardware/software components.

In case of a research problem, how the proposed research solution is expected to be better than the state-of-the-art?

(1 – 2 pages)

Chapter 5

Methodology

Distribute the project goals into smaller objectives/modules and outline deliverables for each objective. Explain the modules of the project through a system-level block diagram. Students may also mention tools, technologies, and suitability of the method(s) to be employed with justification. In case of a research problem, outline the approaches that will be investigated in the project. The proposed work must have a significant design element, open-ended problems are strictly discouraged. The proposed work must also tend to one or more real-world applications. (4 – 6 pages)

Further breakdown of this chapter for the guidance of students is given below.

5.1 Design Constraints

The design constraints must be clearly listed. Care must be taken to ensure that the listed constraints are practical and achievable.

5.2 Technical Approach

Describe the technical strategy to achieve your objectives. Outline the main project phases and mention specific components, software tools, and simulation techniques you plan to use.

5.3 Block Diagram

Insert a clear and well-labeled block diagram here. The diagram should visually represent the main functional blocks of your system and the signal/power flow between them.

5.4 Flow Chart

Insert a flow chart that illustrates the operational logic of your project's software or control system. Use standard symbols to show the process from start to finish, including initialization, data reading, decision points, and actions.

5.5 Applications

Write a comprehensive subsection on the application of the proposed work for solving one or more real-world problems.

Chapter 6

Milestones and Deliverables

Clear milestones should be defined at the start of the project in the form of a Gantt chart. It is recommended to use excel or some equivalent software to make a Gantt chart.

Also describe the final deliverables of your project. Explain the significance of these outcomes.

(1 – 2 pages)

Chapter 7

Work Division

Clear work division among group members must be defined. (1 page)

Chapter 8

Costing

Make a table of major required components with estimated prices. Give a reasonably good estimate of the cost of implementation. (1 page)

Appendix A

Introduction to Latex (Do not include this in the final version of your synopsis)

The material provided in this appendix is taken from
<http://www.sunilpatel.co.uk/thesistemplate.php>

A.1 Learning L^AT_EX

L^AT_EX is not a WYSIWYG (What You See is What You Get) program, unlike word processors such as Microsoft Word or Corel WordPerfect. Instead, a document written for L^AT_EX is actually a simple, plain text file that contains *no formatting*. You tell L^AT_EX how you want the formatting in the finished document by writing in simple commands amongst the text, for example, if I want to use *italic text for emphasis*, I write the ‘\emph{} command and put the text I want in italics in between the curly braces. This means that L^AT_EX is a “mark-up” language, very much like HTML.

A.1.1 A (not so short) Introduction to L^AT_EX

If you are new to L^AT_EX, there is a very good eBook – freely available online as a PDF file – called, “The Not So Short Introduction to L^AT_EX”. The book’s title is typically shortened to just “lshort”. You can download the latest version (as it is occasionally updated) from here:

<http://www.ctan.org/tex-archive/info/lshort/english/lshort.pdf>

It is also available in several other languages. Find yours from the list on this page:

<http://www.ctan.org/tex-archive/info/lshort/>

It is recommended to take a little time out to learn how to use L^AT_EX by creating several, small ‘test’ documents. Making the effort now means you’re not stuck learning the system when what you *really* need to be doing is writing your thesis.

A.1.2 A Short Math Guide for L^AT_EX

If you are writing a technical or mathematical thesis, then you may want to read the document by the AMS (American Mathematical Society) called, “A Short Math Guide for L^AT_EX”. It can be found online here:

<http://www.ams.org/tex/amslatex.html>

under the “Additional Documentation” section towards the bottom of the page.

A.1.3 Common L^AT_EX Math Symbols

There are a multitude of mathematical symbols available for L^AT_EX and it would take a great effort to learn the commands for them all. The most common ones you are likely to use are shown on this page:

<http://www.sunilpatel.co.uk/latexsymbols.html>

You can use this page as a reference or crib sheet, the symbols are rendered as large, high quality images so you can quickly find the L^AT_EX command for the symbol you need.

A.1.4 Figures

There will hopefully be many figures in your thesis (that should be placed in the ‘Figures’ folder). The way to insert figures into your thesis is to use a code template like this:

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width = 1.5in]{./Figures/uet_logo.pdf}
  \rule{35em}{0.5pt}
  \caption{The UET Laore logo.}
  \label{fig:uet_logo}
\end{figure}
```

Also look in the source file. Putting this code into the source file produces the picture of the UET logo that you can see in the figure below.



FIGURE A.1: The UET Lahore logo.

Sometimes figures don’t always appear where you write them in the source. The placement depends on how much space there is on the page for the figure. Sometimes there

is not enough room to fit a figure directly where it should go (in relation to the text) and so L^AT_EX puts it at the top of the next page. Positioning figures is the job of L^AT_EX and so you should only worry about making them look good!

Figures usually should have labels just in case you need to refer to them (such as in figure A.1). The ‘\caption’ command contains two parts, the first part, inside the square brackets is the title that will appear in the ‘List of Figures’, and so should be short. The second part in the curly brackets should contain the longer and more descriptive caption text.

The ‘\rule’ command is optional and simply puts an aesthetic horizontal line below the image. If you do this for one image, do it for all of them.

The L^AT_EX Thesis Template is able to use figures that are either in the PDF or JPEG file format. It is recommended that you read this short guide on how to get the best out of figures in L^AT_EX, available here:

<http://www.sunilpatel.co.uk/texhelp5.html>

Though it is geared more towards users of Mac and OS X systems, much of the advice applies to creating and using figures in general. It also explains why the PDF file format is preferred in figures over JPEG.

A.1.5 Typesetting mathematics

If your thesis is going to contain heavy mathematical content, be sure that L^AT_EX will make it look beautiful, even though it won’t be able to solve the equations for you.

The “Not So Short Introduction to L^AT_EX” (available [here](#)) should tell you everything you need to know for most cases of typesetting mathematics. If you need more information, a much more thorough mathematical guide is available from the AMS called, “A Short Math Guide to L^AT_EX” and can be downloaded from:

<ftp://ftp.ams.org/pub/tex/doc/amsmath/short-math-guide.pdf>

There are many different L^AT_EX symbols to remember, luckily you can find the most common symbols [here](#). You can use the web page as a quick reference or crib sheet and because the symbols are grouped and rendered as high quality images (each with a downloadable PDF), finding the symbol you need is quick and easy.

You can write an equation, which is automatically given an equation number by L^AT_EX like this:

```
\begin{equation}
E = mc^2
\label{eqn:Einstein}
\end{equation}
```

This will produce Einstein's famous energy-matter equivalence equation:

$$E = mc^2 \tag{A.1}$$

All equations you write (which are not in the middle of paragraph text) are automatically given equation numbers by L^AT_EX. If you don't want a particular equation numbered, just put the command, '`\nonumber`' immediately after the equation.

A.2 Sectioning and Subsectioning

You should break your thesis up into nice, bite-sized sections and subsections. L^AT_EX automatically builds a table of Contents by looking at all the '`\chapter{}`', '`\section{}`' and '`\subsection{}`' commands you write in the source.

The table of Contents should only list the sections to three (3) levels. A '`\chapter{}`' is level one (1). A '`\section{}`' is level two (2) and so a '`\subsection{}`' is level three (3). In your thesis it is likely that you will even use a '`\subsubsection{}`', which is level four (4). Adding all these will create an unnecessarily cluttered table of Contents and so you should use the '`\subsubsection*{}`' command instead (note the asterisk). The asterisk (*) tells L^AT_EX to omit listing the subsubsection in the Contents, keeping it clean and tidy.

References